ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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Cost Advantage Analysis

EXECUTIVE SUMMARY

Tobe Energy's market expansion strategy leverages our ultra-efficient membrane-free electrolysis technology in U.S. regions characterized by low electricity costs and exceptionally high prices for delivered hydrogen. This document outlines a phased strategic expansion plan, commencing with an initial flagship facility in Oklahoma City, followed by replication in Tulsa, OK, and subsequent entries into high-potential markets including Washington, New Mexico, South Dakota, and Iowa. The plan leverages a modular facility design, enabling rapid deployment and scalability, targeting industrial, energy, healthcare, and research sectors currently reliant on expensive (often **$20–25+/kg**), logistically constrained hydrogen supplies. This approach positions Tobe Energy for accelerated growth, market leadership, and offers a compelling investment opportunity in the burgeoning green hydrogen economy.

1. INTRODUCTION: THE TOBE ENERGY VALUE PROPOSITION

Tobe Energy has developed an advanced, ultra-efficient membrane-free electrolysis process capable of producing green hydrogen at costs significantly below current market rates for delivered hydrogen, particularly in regions underserved by existing production infrastructure. Our mission is to make green hydrogen economically competitive, thereby accelerating the energy transition. This market expansion plan details a pragmatic, phased approach to establishing a network of decentralized hydrogen production facilities across strategically selected U.S. markets where incumbent delivered hydrogen costs are exceedingly high.

Key differentiators:

- **>92% system efficiency** (HHV) and **94.7% stack efficiency** (LHV) for our membrane-free electrolyzers
- **$1.81/kg** levelized cost of hydrogen (LCOH) at optimal operating conditions
- Modular design enabling rapid deployment and scalability
- Targeting markets where delivered hydrogen currently costs **$20–25+/kg**

2. PHASE 1: INITIAL MARKET ENTRY & VALIDATION — OKLAHOMA CITY, OK (YEAR 1)

Strategic Rationale

Oklahoma City (OKC) has been selected as the optimal launch market due to a compelling confluence of factors:

- **Low-Cost Electricity**: Industrial electricity rates average approximately **6.3¢/kWh**

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- **Significant Renewable Penetration**: Approximately **35%** of Oklahoma's electricity is wind-generated, offering a pathway to green hydrogen
- **Incumbent High Hydrogen Prices**: Local industries, particularly smaller volume or specialized users, currently pay premium prices, often in the range of **$20–25+/kg**, for hydrogen trucked in from distant production hubs or supplied in cylinders

Target Customer Segments

The initial OKC facility will serve a diversified customer base with established hydrogen demand:

Industrial/Manufacturing

- Bodycote Thermal Processing (metal heat treatment)
- Current hydrogen consumption: ~150 kg/day at $21.50/kg

Energy Technology & Utilities

- Baker Hughes Energy Innovation Center (R&D for hydrogen-blended turbines)
- Oklahoma Gas & Electric (OG&E) (generator cooling)
- OG&E operates 4 power plants using hydrogen cooling, consuming ~500 kg/week

Healthcare & Research

- Oklahoma Medical Research Foundation (OMRF)
- OU Health
- Integris Baptist Medical Center
- Mercy Hospital (analytical instrumentation, medical research)
- Regional medical facilities consume ~300 kg/week of high-purity hydrogen

Objectives

- Validate Tobe Energy's production cost model and operational efficiency at commercial scale
- Secure anchor off-take agreements, demonstrating market demand and de-risking further investment by offering substantial savings against current delivered prices

- Establish a replicable operational blueprint for subsequent facilities
- Achieve 70% capacity utilization by end of Year 1

3. PHASE 2: REGIONAL EXPANSION — TULSA, OK (YEAR 2)

Strategic Rationale

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Tulsa mirrors OKC's favorable market conditions, including low-cost electricity, access to wind power, and reliance on expensive delivered hydrogen (often **$20–25+/kg** for many users). This allows for rapid replication of the validated OKC model within the same supportive state regulatory environment.

- **Electricity Costs**: Industrial rates average **6.1¢/kWh**
- **Renewable Energy**: PSO (Tulsa's utility) has **37% wind power capacity**
- **Hydrogen Demand**: Similar industrial and healthcare sectors as OKC with established hydrogen use

Target Customer Segments

Manufacturing & Engineering

- John Zink Hamworthy Combustion
- Zeeco Inc. (combustion systems testing & development)
- Linde Engineering (hydrogen solutions R&D)
- Combined estimated consumption: ~200 kg/day

Energy Utilities

- Public Service Company of Oklahoma (PSO) (generator cooling)
- PSO uses ~750 kg/week for generator cooling

Healthcare & Education

- OSU Medical Center
- Saint Francis Hospital
- University of Tulsa (laboratory and research applications)
- Tulsa healthcare sector consumes ~350 kg/week

Objectives

- Expand production capacity to meet growing regional demand
- Refine deployment processes and achieve economies of scale in regional operations
- Strengthen Tobe Energy's market presence within Oklahoma by displacing high-cost incumbent hydrogen

- Achieve 80% capacity utilization across OK operations by end of Year 2

4. PHASE 3: STRATEGIC OUT-OF-STATE ENTRY — PACIFIC NORTHWEST (WASHINGTON) (YEARS 3–4)

Strategic Rationale

Washington state presents a prime expansion opportunity due to:

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- **Nationally Low Electricity Costs**: Industrial rates average approximately **6.35¢/kWh**, primarily from abundant, renewable hydropower
- **High Delivered Hydrogen Prices**: Significant dependence on imported hydrogen leads to current costs often reaching **$20–25+/kg** for distributed consumers
- **Strong Green Energy Profile**: Ideal for producing deeply decarbonized hydrogen with **95% renewable grid mix**

Target Customer Segments

Focus on industrial users (e.g., advanced manufacturing, food processing), research institutions, and emerging mobility applications that currently face exceedingly high hydrogen procurement costs.

- Washington state hydrogen market size: ~2,500 kg/day
- Key growth sectors: semiconductor manufacturing (15% growth), green steel (new projects)

Objectives

- Demonstrate the portability and adaptability of Tobe Energy's modular production model to a new regulatory and market environment
- Capture market share by significantly undercutting incumbent delivered hydrogen prices
- Establish a foothold in the Pacific Northwest, a region with strong sustainability goals
- Achieve 60% market penetration in target industrial segments within 18 months of operation

5. PHASE 4: BROADER U.S. EXPANSION — TARGETING UNDERSERVED MARKETS (YEARS 4–6)

Based on the success of initial phases, further expansion will target states with similar favorable characteristics: low electricity costs, high renewable energy potential, and current reliance on expensive, imported hydrogen (where delivered prices for many users can reach **$20–25+/kg**). Key identified states include:

New Mexico

- Characterized by some of the nation's lowest industrial electricity rates (~**5.75¢/kWh**)
- High renewable penetration (primarily wind and solar, ~35% of generation)
- Significant hydrogen import dependency
- Emerging hydrogen hub development with $150M in federal funding

South Dakota

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- Offers abundant wind energy resources (contributing ~**56% of in-state generation**)
- Currently has no local large-scale hydrogen production, leading to high delivered costs
- Hydrogen demand growth rate: 12% annually

Iowa

- Leads the nation in wind energy penetration (~**60% of electricity mix**)
- Relies on externally sourced hydrogen for industrial and emerging uses, with associated high delivered prices
- Major manufacturing base with potential for hydrogen integration

Each new market entry will replicate the proven modular facility deployment strategy, targeting local industrial, healthcare, research, and emerging mobility hydrogen consumers currently facing very high hydrogen costs.

6. BUSINESS MODEL AND COMPETITIVE ADVANTAGE

Tobe Energy's market expansion is underpinned by a robust business model and distinct competitive advantages:

Disruptive Cost Efficiency

- Our core value proposition is the ability to produce green hydrogen on-site at a cost significantly lower than current delivered prices (often **$20–25+/kg** for target customers) by leveraging low-cost regional electricity (targeting operations with electricity at ~6¢/kWh or less) and superior electrolysis efficiency
- Current market comparison: Tobe LCOH of **$1.81/kg** vs. delivered hydrogen at $22.50/kg = **92% cost reduction**

Modular and Scalable Deployment

- Standardized, containerized, or skid-mounted electrolysis units enable rapid deployment, minimize upfront capital expenditure per site, and allow for flexible scaling in response to local demand

- Production facilities will be strategically located near end-users to minimize transportation costs
- Deployment timeline: 6–8 months from site selection to operation

Customer-Centric Market Entry

- Initial operations in each new market will be anchored by securing offtake agreements with key local hydrogen consumers
- Ensuring baseline demand, validating market viability, and providing revenue predictability based on substantial cost savings for these customers
- Typical anchor customer commitment: 5-year contracts at 25–30% below current market rates

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7. FINANCIAL STRATEGY & INVESTMENT PATHWAY

A phased fundraising approach will fuel Tobe Energy's expansion:

Initial Capital (Seed/Pilot Phase)

- Funding for the establishment and initial operation of the Oklahoma City facility, focused on demonstrating technological viability, achieving target production costs, and securing initial customer contracts based on compelling price advantages
- Seed round: **$10M** targeting 500 kg/day capacity in OKC
- Projected revenue Year 1: **$3.2M** at 70% capacity utilization

Growth Capital (Series A and subsequent rounds)

- Upon successful validation in Oklahoma, subsequent funding rounds will finance the replication of facilities across Tulsa and into the identified expansion states (Washington, New Mexico, South Dakota, Iowa, etc.)
- Each funding tranche will be tied to achieving specific deployment milestones and revenue targets
- Series A target: **$25M** for regional expansion (Year 2–3)
- Projected revenue Year 3: **$18.5M** across multiple facilities

8. CONCLUSION AND INVESTMENT OPPORTUNITY

Tobe Energy's strategic market expansion plan targets a clear and substantial market inefficiency: regions with access to low-cost (often renewable) electricity that are simultaneously burdened by exceptionally high prices for delivered hydrogen (often **$20–25+/kg** for many end-users). Our proprietary, ultra-efficient membrane-free electrolysis technology, combined with a pragmatic, modular deployment strategy, positions Tobe Energy for rapid growth and market leadership in these underserved areas.

The U.S. hydrogen market is projected to grow from **$3.8B in 2023** to **$11.2B by 2027** (CAGR 30.7%), with green hydrogen representing the fastest-growing segment.

This plan offers investors a clear pathway to participate in a scalable, profitable, and sustainable venture that is set to transform the accessibility and economics of green hydrogen. Tobe Energy is not just building hydrogen plants; we are enabling regional energy transitions and creating significant value for our customers and stakeholders by providing a far more economical hydrogen supply.

- **ROI Projection**: 5.2x return by Year 5 based on current market conditions
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- **Market Capture**: Targeting 15% of regional hydrogen markets within 5 years
- **Scalability**: Model designed for 20+ facility replication across U.S. markets
- **Exit Potential**: Strategic acquisition interest from major energy players at 10–12x revenue multiple

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